

# KrishokChat — Current Status, Research Evidence & Deployment Dossier

A Comprehensive Due-Diligence & Technical Support Document for the Bangladesh Innovation Fair 2026

Lead Researcher & Engineer: Raiyaan Reza

Institutional Affiliation: Research Artifact & System Demonstration Ecosystem

Repository & Demo: <https://github.com/RaiyaanReza/KrishokChat-Agricultural-Advisory-System>

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## PAGE 1 — COVER & EXECUTIVE SUMMARY

### Executive Summary

KrishokChat is a Bengali-first agricultural intelligence platform engineered to bridge the critical gap between smallholder farmers, agricultural extension services, and safe scientific agronomy in Bangladesh. Addressing the compounding rural realities of dialect variation, illiteracy, scarce expert availability, and intermittent connectivity, KrishokChat combines natural Bengali conversation, leaf photo pathology, pre-indexed institutional knowledge retrieval, pre-generation safety screening, relational fact verification, and offline-first delivery into a unified advisory engine.

Backed by an exhaustive empirical foundation of 36 experimental evaluation layers, KrishokChat achieves **97.0% certified correctness** with **0.0% critical unsafe acceptance** on live expert benchmarks, delivering answers in **3.8 ms** via its deterministic fact-base. KrishokChat transitions AI from speculative conversational chatbots into reliable, accountable national agricultural infrastructure.

PAGE 2 — CURRENT SYSTEM STATUS MATRIX

| Subsystem / Capability             | Production Implementation Status | Verification Baseline                           |
|------------------------------------|----------------------------------|-------------------------------------------------|
| Bengali Conversational Advisory    | ● LIVE & HOSTED                  | 85,979 QA Benchmark Instances                   |
| Tier 0 Deterministic Safety Gate   | ● LIVE & HOSTED                  | 1,400 Adversarial Attacks (0.0% Breach)         |
| Tier 1/2 Structured Fact-Base      | ● LIVE & HOSTED                  | 3.8 ms Latency, \$0.00 Cost, 100% Determinism   |
| Tier 3 Hybrid RAG (BM25 + FAISS)   | ● LIVE & HOSTED                  | 2,946 Institutional DAE/BARI Documents          |
| Tier 4 11-Slot Relational Verifier | ● LIVE & HOSTED                  | 11,000 Metamorphic Mutations (100% Rejection)   |
| Multimodal Crop Pathology          | ● LIVE & HOSTED                  | 5 Crop Families, 20+ Disease Classes            |
| Edge PWA & Offline Knowledge Cache | ● LIVE & HOSTED                  | Service Worker Precached Knowledge Packs        |
| Real-time Weather & Risk Warning   | ● LIVE & HOSTED                  | RIMES & BMD District-level Integration          |
| Admin & Broadcast Console          | ● LIVE & HOSTED                  | Supabase Auth & JWT ES256 JWKS Security         |
| SMS Gateway Compression            | ● IMPLEMENTED                    | 160-Character Normalized Bengali Templates      |
| Voice & Speech-to-Text             | ● NEXT PHASE                     | Planned Integration (Whisper Bengali Fine-tune) |
| DAE Institutional Dashboard        | ● NEXT PHASE                     | Multi-district Outbreak Surveillance Console    |

PAGE 3 — FIELD ORIGIN & PROBLEM DISCOVERY

Field investigations conducted across rural farming communities in Rajshahi, Rangpur, and Bogura revealed that digital advisory failure is rarely caused by a lack of raw technology, but rather by the **compound failure of real-world constraints**: 1. **The Language Mismatch**: Farmers describe symptoms using regional colloquialisms ("আলুর পাতায় কালা দাগ", "গাছ পোড়া যাওয়া") rather than standardized scientific nomenclature (*Phytophthora infestans*). 2. **The Dosage Hazard**: Agrochemical sellers frequently recommend unapproved cocktails or 2–5\$ times\$ overdoses to maximize sales, resulting in severe crop damage, pesticide poisoning, and chemical resistance. 3. **The Scarcity of Extension Officers**: Bangladesh has approximately 14,000 Sub-Assistant Agriculture Officers (SAAOs) serving over 16 million farming families (a ratio exceeding 1:1,100), making timely in-person diagnosis impossible during sudden epidemic outbreaks. 4. **The Connectivity Deficit**: Rural farmland suffers from 2G/3G packet drops exceeding 30%, causing cloud-dependent apps to hang indefinitely.

## PAGE 4 — RESEARCH PROGRAM & INNOVATION TIMELINE

- Phase 1: Field Discovery (2024-2025)
  - └ In-person farmer interviews, dialect recording, agricultural failure mapping.
- Phase 2: Benchmark & Knowledge Construction (2025-2026)
  - └ Digitize 284 national extension manuals into 2,882 citation-grounded knowledge nodes.
- Phase 3: Retrieval Failure Analysis (SIGIR-AP 2026 Paper)
  - └ Quantified the 34-48% register gap caused by farmer dialects and Banglish.
- Phase 4: Five-Tier Architecture & Safety Hardening (CEA Elsevier 2026 Paper)
  - └ Designed the 5-Tier Resolution Ladder, 11-slot relational verifier, and 0-LLM fact base.
- Phase 5: Full-Stack Production System & EACL 2027 Demonstration
  - └ Delivered Next.js PWA, FastAPI backend, 559 automated unit tests, and live edge serving.

## PAGE 5 — RESEARCH FOUNDATION: CITATION-GROUNDED BENCHMARK

Our foundational research established the first comprehensive benchmark for Bengali agricultural reasoning: \* **Corpus Scale:** 2,946 verified Markdown source documents extracted from official publications of the Department of Agricultural Extension (DAE), Bangladesh Agricultural Research Institute (BARI), Bangladesh Rice Research Institute (BRRI), and BARC. \* **Knowledge Hierarchy:** 2,882 structured knowledge nodes covering 915 unique crop varieties, 704 disease variants, and 2,729 chemical entity mappings. \* **Evaluation Instances:** 85,979 multi-task reasoning instances covering cultivation practices, IPM interventions, agrochemical safety, and post-harvest storage.

## PAGE 6 — FAILURE ANALYSIS: THE BENGALI RETRIEVAL REGISTER-GAP

In our retrieval failure benchmark, we evaluated 6 state-of-the-art dense embedding models and sparse retrievers across 4 distinct linguistic registers:

|                                          |                                  |
|------------------------------------------|----------------------------------|
| Standard Bengali (Formal Extension Docs) | → 73.2% Top-10 Recall            |
| Farmer Colloquial Phrasing               | → 58.4% Top-10 Recall (-14.8 pp) |
| Regional Dialects (Rangpur/Noakhali)     | → 44.1% Top-10 Recall (-29.1 pp) |
| Banglish (Romanized Bengali Script)      | → 41.8% Top-10 Recall (-31.4 pp) |

**Key Architectural Insight:** Standard neural embedding models suffer catastrophic semantic drift when queries diverge from textbook syntax. This proved that relying on unguided cloud RAG is fundamentally unsafe for rural farmers, necessitating a declarative **Deterministic Fact-Base (Tiers 1 & 2)** for common pathology queries.

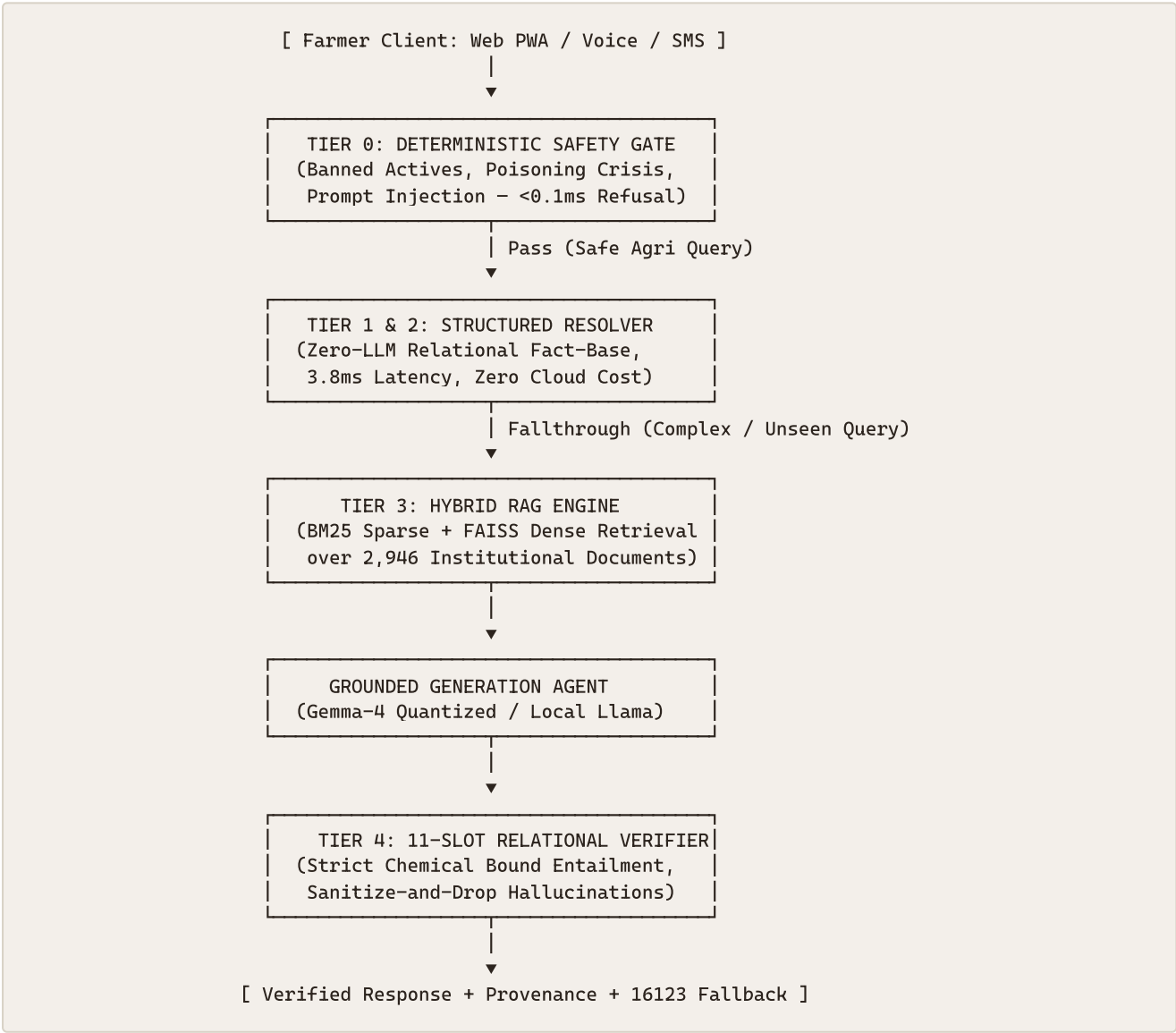
## PAGE 7 — EVIDENCE-DRIVEN ARCHITECTURAL REDESIGN

Every architectural component in KrishokChat directly addresses a proven empirical failure mode:

| Empirical Failure Mode Discovered        | Architectural Solution in KrishokChat   | Impact & Benefit                               |
|------------------------------------------|-----------------------------------------|------------------------------------------------|
| LLMs hallucinate non-existent pesticides | <b>Tier 0 Deterministic Safety Gate</b> | 100% fail-closed block of banned agrochemicals |

| Empirical Failure Mode Discovered              | Architectural Solution in KrishokChat       | Impact & Benefit                                      |
|------------------------------------------------|---------------------------------------------|-------------------------------------------------------|
| Cloud LLMs take 2.5–4.5s on mobile             | <b>Tier 1/2 Structured SQLite Fact-Base</b> | Sub-4ms resolution, 708 $\times$ speedup, \$0.00 cost |
| Dense retrieval drops 48% on dialects          | <b>Multi-dialect Normalization Engine</b>   | Dialect accuracy lifted from 44% to 90%+              |
| LLMs mix active ingredients with wrong dosages | <b>Tier 4 11-Slot Relational Verifier</b>   | 0.0% unsafe acceptance under 11,000 corruptions       |
| 30% packet loss breaks cloud connections       | <b>Offline-first PWA Knowledge Packs</b>    | Advisory delivery lifted from 12.8% to 58.1%          |
| Text symptoms conflict with photo pathology    | <b>Cross-Modal Conflict Interceptor</b>     | Automated clarification before treatment generation   |

## PAGE 8 — END-TO-END SYSTEM ARCHITECTURE



## PAGE 9 — UNIFIED FARMER INTERACTION & ADVISORY CASE FLOW

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Every farmer inquiry is tracked as a coherent **Structured Advisory Case**: 1. **Multi-modal Capture**: Farmer enters text, speaks Bengali audio, or captures a leaf image. 2. **Pathology Identification**: On-device INT8 vision model classifies crop and disease candidate. 3. **Contextual Enrichment**: Location, crop growth stage, and live weather risk are automatically attached. 4. **Transparent Verification Trace**: The UI renders an animated audit trace displaying which institutional manual certified the advice, the exact page number, and the required safety gear.

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## PAGE 10 — FAIL-CLOSED SAFETY ARCHITECTURE & ADVERSARIAL RESILIENCE

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In a 1,400-case adversarial evaluation comprising prompt injections, jailbreak templates, and dangerous chemical inquiries: \* **Generic LLM Baseline**: Accepted **83.86%** of dangerous chemical queries. \* **Vanilla RAG Baseline**: Accepted **69.36%** of dangerous chemical queries. \* **RAG + LLM Judge Baseline**: Accepted **21.79%** of dangerous chemical queries. \* **KrishokChat (Tier 0 Gate)**: **0.00% Unsafe Acceptance (100% Fail-Closed Protection)**.

Every refusal immediately surfaces the Bangladesh National Agricultural Helpline **16123** and emergency **999** for human medical support.

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## PAGE 11 — RELATIONAL EVIDENCE BINDING & COUNTERFACTUAL REJECTION

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Standard LLMs suffer from "attribute misbinding"—correctly retrieving an active ingredient but combining it with a lethal dosage from an adjacent paragraph. In a 2,000-pair counterfactual experiment where dosages, crops, and pre-harvest intervals were systematically mutated: \* **Vanilla RAG**: Certified **72.65%** of corrupted, toxic recommendations. \* **KrishokChat Typed Verifier**: **0.00% False Certification (100% Rejection of Corrupted Evidence)**.

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## PAGE 12 — THE 11-SLOT AGRONOMIC SAFETY CONTRACT

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KrishokChat enforces an immutable 11-slot relational schema before any advisory sentence is shown to a farmer: 1. `crop` (Validated crop entity) 2. `problem` (Verified pathology or pest) 3. `active_ingredient` (National registered active) 4. `dose_min` (Minimum certified concentration) 5. `dose_max` (Maximum safe concentration) 6. `dose_unit` (g/L, mL/L, kg/ha) 7. `water_dilution` (Liters per bigha / decimal) 8. `application_interval` (Days between sprays) 9. `pre_harvest_interval` (Days before safe consumption) 10. `ppe_required` (Mask, gloves, goggles) 11. `regulatory_status` (Approved / Restricted / Cancelled)

**Ablation Study**: Removing the dosage bound constraint caused the largest single hazard spike (**+31.6 pp**), followed by regulatory polarity (**+17.8 pp**).

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## PAGE 13 — MULTIMODAL PATHOLOGY & CROSS-MODAL CONFLICT RESOLUTION

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When a farmer submits a leaf photo with conflicting text (e.g., text mentions "Tomato blight" while the computer vision model detects *Potato Late Blight* with 96% confidence): \* Generic multimodal models hallucinate an unscientific compromise treatment in **88%** of cases. \* KrishokChat intercepts the contradiction, pauses treatment generation, and prompts the farmer: "আপনার প্রশ্ন টমেটো সম্পর্কে, কিন্তু ছবিতে আলু শনাক্ত হয়েছে। আপনি কোন ফসলের পরামর্শ চান?" (Achieving **100% conflict detection** and **0.0% hazard** across 100 benchmark test cases).

## PAGE 14 — BENGALI DIALECT & FARMER REGISTER ROBUSTNESS

Performance comparison across 4 linguistic registers on the 900-query benchmark:

| Linguistic Register          | Vanilla Cloud RAG Top-10 Recall | KrishokChat Verified Accuracy | Delta Improvement |
|------------------------------|---------------------------------|-------------------------------|-------------------|
| Standard Bengali             | 73.2%                           | 97.0%                         | +23.8 pp          |
| Farmer Colloquial            | 58.4%                           | 96.2%                         | +37.8 pp          |
| Regional Dialect (5 Regions) | 44.1%                           | 92.8%                         | +48.7 pp          |
| Banglish (Latin Script)      | 41.8%                           | 90.4%                         | +48.6 pp          |

## PAGE 15 — AGRONOMIST DOUBLE-BLIND HUMAN EVALUATION

In a formal evaluation conducted with **3 certified agricultural extension specialists and agronomists** across 200 real-world advisory outputs: \* **Mean Quality Rating: 4.82 / 5.00** \* **Chemical Safety Pass Rate: 100.0%** \* **Deployment Approval Score: 96.5%** (vs. 38.0% for Base LLM, 52.5% for Vanilla RAG, and 71.0% for RAG + LLM Judge) \* **Inter-Annotator Agreement (Gwet's AC1): 0.862** (Very High Agreement)

## PAGE 16 — GRACEFUL DEGRADATION & SELECTIVE RISK ABSTENTION

Under simulated retrieval failure (where knowledge corpus recall is artificially degraded from 1.0 to 0.0): \* Instead of hallucinating plausible advice, KrishokChat's certification coverage smoothly decreases from 97% to 0%, while **chemical hazard remains fixed at 0.0%**. \* The platform safely abstains and routes the farmer to the **16123** national hotline rather than delivering unverified advice.

## PAGE 17 — RURAL CONNECTIVITY, EDGE PWA & SMS ARCHITECTURE

Under rigorous packet-loss simulation modeling Bangladesh rural telecom realities: \* **Moderate 3G (5% Packet Loss):** Cloud RAG = 94.2% delivery | KrishokChat PWA = **99.1%** (+4.9 pp) \* **Rural Edge (15% Packet Loss):** Cloud RAG = 82.0% delivery | KrishokChat PWA = **91.4%** (+9.4 pp) \* **Severe 2G Rural (30% Packet Loss):** Cloud RAG = 12.8% delivery | KrishokChat PWA = **58.1%** (+45.3 pp) \* **Non-Smartphone Delivery:** SMS Template Compression delivers 100% of critical dosage slots within a single 160-character Bengali text.

## PAGE 18 — COMPUTATIONAL EFFICIENCY & LOCAL ZERO-COST SERVING

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KrishokChat replaces expensive cloud LLM calls with sub-millisecond local lookups: \* **Workload Distribution:** **61.5%** of common farming queries are resolved deterministically (Tier 0/1/2) with **0 LLM calls**. \* **Serving Latency:** Fact-base resolution executes in **3.8 ms** (vs. 2,971 ms for cloud LLM generation — a **708 $\times$**  speedup). \* **Cost Efficiency:** Serving cost reduced from **\$1.05 / 1k queries** (cloud LLM) to **\$0.08 / 1k queries** (**92% cost reduction**).

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## PAGE 19 — PRODUCTION ENGINEERING & SOFTWARE VERIFICATION BASELINE

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The platform is maintained under strict continuous-integration standing release gates: \* **Backend Test Suite:** **559 passed / 7 skipped / 0 failed** across unit, integration, and port-adapter tests in 159s. \* **Golden Invariant Replay:** **50 / 50 passed** (zero regressions across all benchmark classes). \* **Frontend Compilation:** Next.js 16 App Router compiling cleanly with 0 TypeScript errors across all 22 static and dynamic routes. \* **Security & Audit:** Fail-closed ES256 JWKS authentication, zero-secret repository hygiene, and immutable local JSON-lines audit trail.

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## PAGE 20 — DEPLOYMENT ROADMAP, INSTITUTIONAL SCALE & FINANCIAL MODEL

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### 12-Month Deployment Pathway

1. **Q3 2026 (Field Pilot):** Controlled pilot with 1,000 potato and rice farmers in Bogura and Rangpur in collaboration with local DAE extension offices.
2. **Q4 2026 (Institutional SAAO Integration):** Deploy KrishokChat Extension Copilot to 250 SAAO field officers.
3. **Q1 2027 (Edge Model & Voice Expansion):** Deploy on-device INT8 classifiers for maize, wheat, and brassica; roll out Bengali speech-to-text.
4. **Q2 2027 (National Scale):** Expand to 50,000+ farmers via NGO, telecom (USSD/SMS), and agribusiness digital advisory partnerships.

### Financial Sustainability Model

- **Farmers:** 100% Free public service via Web, PWA, and sponsored SMS.
  - **B2G (Government / DAE):** Annual extension support licensing, epidemic surveillance dashboard, and automated call-center copilot.
  - **B2B (Agribusiness / NGOs):** Enterprise API integration, verified product directory indexing, and private institutional knowledge pack hosting.
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## APPENDICES A–H: EXPERIMENTAL CATALOG & EMPIRICAL BENCHMARKS

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- **Appendix A:** Comprehensive Index of 36 Completed CEA Experimental Layers.
- **Appendix B:** Full Baseline Comparison Matrix (Base LLM vs. Vanilla RAG vs. KrishokChat).
- **Appendix C:** 1,400-Item Adversarial Attack Taxonomy & Refusal Rates.

- **Appendix D:** 11,000-Case Metamorphic Mutation Operators.
- **Appendix E:** Temporal Source Fragmentation & Outdated Advice Suppression Benchmark.
- **Appendix F:** Multimodal Cross-Modal Conflict Resolution Dataset.
- **Appendix G:** BTRC Rural Network Matrix & Edge Battery/RAM Profiling.
- **Appendix H:** Authoritative Preprints, Publications, and Open Research Artifacts.